

K2-29b

R-2364

Benchmark run · workflow W8-benchmark-deep · record deep-bench_K2-29_250917 · created 2026-08-19T23:45:15+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460935.9132 +/- 0.0083 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.171 +/- 0.05
Transit depth (Rp/Rs)^2	2.92 +/- 1.72 [%]
Orbital Inclination (inc)	85.13 +/- 2.39
Airmass coefficient 1 (a1)	0.96 +/- 0.13
Airmass coefficient 2 (a2)	0.05 +/- 0.29
Scatter in the residuals of the lightcurve fit is	14.17 %
Best Comparison Star	#1 - [392, 242]
Optimal Method	PSF photometry
Transit Duration (day)	0.051 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.171 $\pm$ 0.05 vs 0.1419 (deviation 0.58 σ)
Duration fit vs published	0.051 d vs 0.0925 d (deviation 1.34 σ)
Mid-time O–C	-19.2 min
Depth SNR	3.4
Residual scatter	14.17 %

Light curve

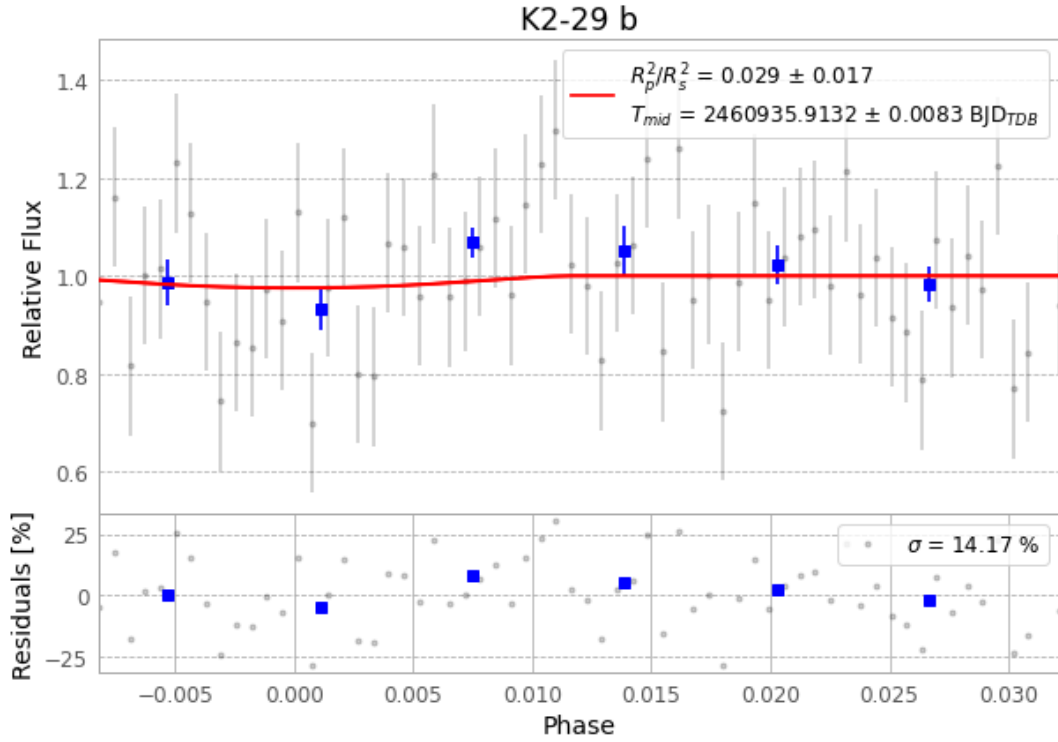


Figure 1 — FinalLightCurve_K2-29 b_2025-09-17.png (SHA-256 f558aaabd09716f8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [346, 64], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 e666810e4d756f71...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

63 FITS frames (41 MB), K2-29250917091215.FITS → K2-29250917122118.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-K2-29-250917.log (d0ad3920a542...), FinalLightCurve_K2-29 b_2025-09-17.png (f558aaabd097...), FinalLightCurve_K2-29 b_2025-09-17.csv (2914f16e9b8a...)

Compute resources

Wall time 421.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-19 23:45Z.